LOCAL FREENESS DETECTED BY THE FITTING IDEAL OF A PRESENTATION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Fitting ideal of a finite module, defined through the minors of a presentation matrix. defined in the same section, source lines 1320-1354, before the counted statement. Supplies the ideal whose relation to local freeness is asserted.
- Criterion for local freeness of a finite module over a ring through the rank function and the Fitting ideals. standard, used at proof lines 1380-1400. Converts the rank condition into the statement that a minor generates the unit ideal.
- Behaviour of the maximal minors of a matrix under localisation. used in both directions of the proof, lines 1370-1410. Lets the argument be carried out after localising at a prime.

Source reference. Original lemma 07Z8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10138–10138. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07Z7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10137–10137. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00O3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 881–881. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let M be a finite R-module. Choose a presentation*

$$\bigoplus\nolimits_{j\in J} R \longrightarrow R^{\oplus n} \longrightarrow M \longrightarrow 0.$$

of M. Let $A = (a_{ij})_{i=1,\ldots,n, j\in J}$ be the matrix of the map $\bigoplus_{j\in J} R \to R^{\oplus n}$. The ideal $Fit_k(M)$ generated by the $(n-k)\times(n-k)$ minors of A is independent of the choice of the presentation.

Proof. Let $K \subset R^{\oplus n}$ be the kernel of the surjection $R^{\oplus n} \to M$. Pick $z_1, \ldots, z_{n-k} \in K$ and write $z_j = (z_{1j}, \ldots, z_{nj})$. Another description of the ideal $\mathrm{Fit}_k(M)$ is that it is the ideal generated by the $(n-k)\times(n-k)$ minors of all the matrices (z_{ij}) we obtain in this way.

Suppose we change the surjection into the surjection $R^{\oplus n+n'} \to M$ with kernel K' where we use the original map on the first n standard basis elements of $R^{\oplus n+n'}$ and 0 on the last n' basis vectors. Then the corresponding ideals are the same. Namely, if $z_1, \ldots, z_{n-k} \in K$ as above, let $z'_j = (z_{1j}, \ldots, z_{nj}, 0, \ldots, 0) \in K'$ for $j = 1, \ldots, n-k$ and $z'_{n+j'} = (0, \ldots, 0, 1, 0, \ldots, 0) \in K'$. Then we see that the ideal of $(n-k)\times(n-k)$ minors of (z_{ij}) agrees with the ideal of $(n+n'-k)\times(n+n'-k)$ minors of (z'_{ij}). This gives one of the inclusions. Conversely, given $z'_1, \ldots, z'_{n+n'-k}$ in K' we can project these to $R^{\oplus n}$ to get $z_1, \ldots, z_{n+n'-k}$ in K. By Lemma 07Z7 we see that the ideal generated by the $(n+n'-k)\times(n+n'-k)$ minors of (z'_{ij}) is contained in the ideal generated by the $(n-k)\times(n-k)$ minors of (z_{ij}). This gives the other inclusion.

Let $R^{\oplus m} \to M$ be another surjection with kernel L. By Schanuel's lemma (Algebra, Lemma 00O3) and the results of the previous paragraph, we may assume $m = n$ and that there is an isomorphism $R^{\oplus n} \to R^{\oplus m}$ commuting with the surjections to M. Let $C = (c_{li})$ be the (invertible) matrix of this map (it is a square matrix as $n = m$). Then given $z'_1, \ldots, z'_{n-k} \in L$ as above we can find $z_1, \ldots, z_{n-k} \in K$ with $z'_1 = Cz_1, \ldots, z'_{n-k} = Cz_{n-k}$. By Lemma 07Z7 we get one of the inclusions. By symmetry we get the other. □